Centre Number	Candidate Number	Candidate Name

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ORDINARY LEVEL **6116/2**

PAPER 2 1 hour 30 minutes

Marks 80 **2020**

Additional Material: Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.

- Answer **all** questions.

- The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
Total	

Marker	
Checker	

This document consists of **18** printed pages and **2** blank pages.

Republic of Namibia

MINISTRY OF EDUCATION, ARTS AND CULTURE

1 Namibia is faced with a high poaching rate of black rhinoceros. Their population is decreasing drastically, causing the fear of extinction.

Fig. 1.1 shows a black rhinoceros, *Diceros bicornis*.

Fig. 1.1

(a) State the genus and species of *Diceros bicornis*.

genus ....

species .... [1]

(b) Apart from poaching, suggest and explain any other two possible factors which may cause the rhinoceros to be endangered.

1 ....

....

....

....

2 ....

....

....

.... [4]

For Examiner's Use

(c) Discuss how the conservation of rhinoceros could benefit the Namibian tourism industry.

....

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.... [3]

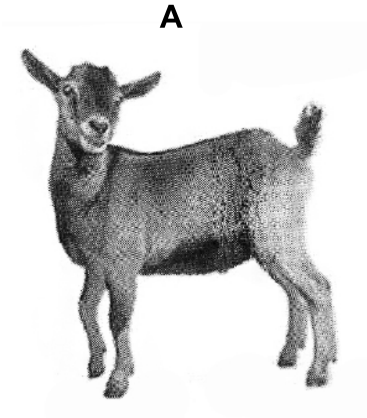
[8]

For Examiner's Use

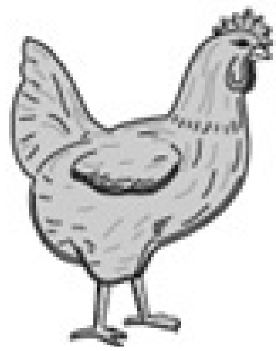
2 Fig. 2.1 shows four vertebrates.

(a) Construct a simple key to identify the organisms in Fig 2.1.

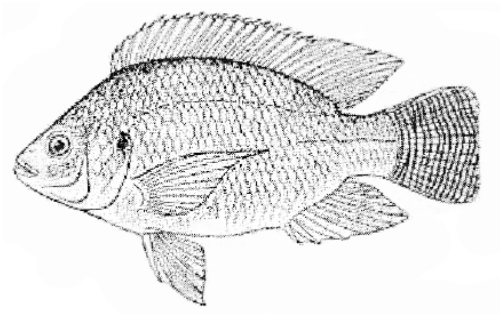
A



B



A



B

Fig. 2.1

[4]

For Examiner's Use

(b) Fig. 2.2 shows the general structure of a virus.

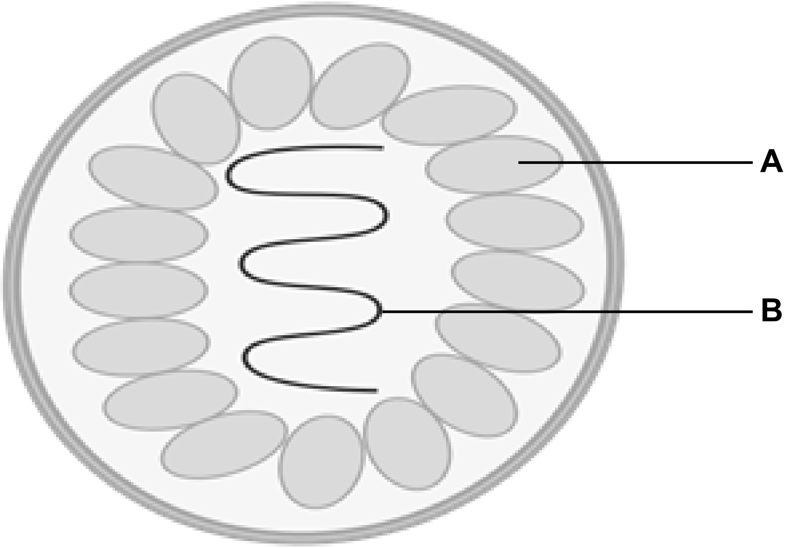


Fig. 2.2

(i) Identify the parts labelled **A** and **B**.

A..

B.. [2]

(ii) Explain the features of viruses that make it difficult to classify them as living organisms.

....

....

....

.... [2]

[8]

For Examiner's Use

3 Fig. 3.1 shows a cross section of a dicotyledonous root

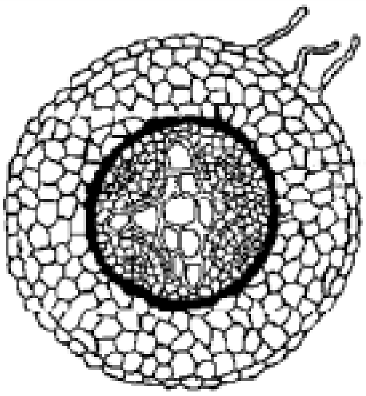


Fig. 3.1

(a) On Fig. 3.1, label the **root hair**, **phloem** and **xylem vessel**. [2]

(b) Mineral salts are absorbed from the soil by the root hairs through active transport.

(i) Define the term *active transport*.

....

....

....

.... [2]

(ii) Suggest the reasons why a root hair cell contains many mitochondria.

....

....

....

.... [2]

[6]

For Examiner's Use

4 Fig. 4.1 shows the female reproductive system

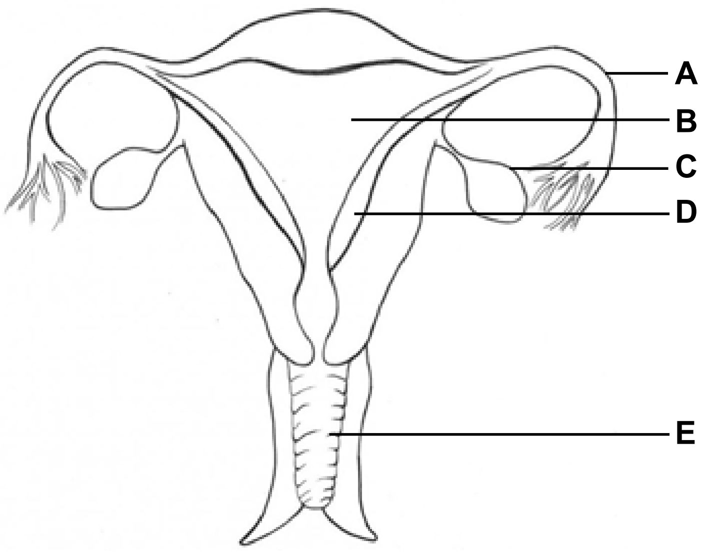


Fig. 4.1

(a) (i) Identify the parts labelled **B**, **C**, **D** and **E**.

B..

C..

D..

E.. [4]

(ii) State the function of the part labelled **A**.

.... [1]

For Examiner's Use

(b) The menstrual cycle is controlled by a number of hormones.

Fig. 4.2 shows levels of two hormones over twenty-eight days.

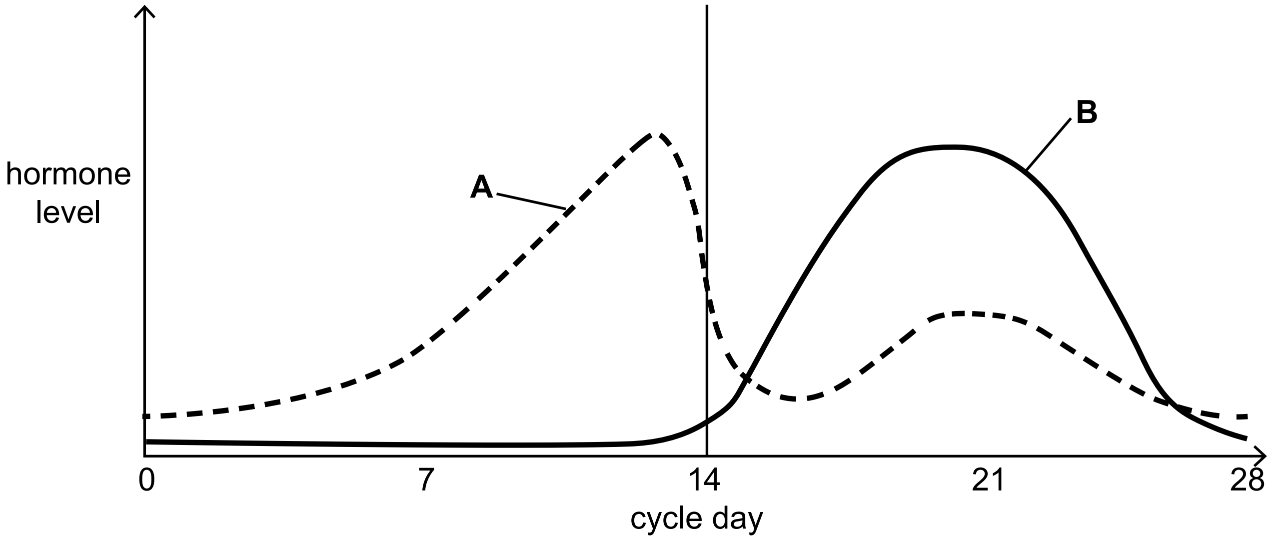


Fig. 4.2

(i) Identify hormones **A** and **B**.

A...

B... [2]

(ii) With reference to Fig.4.2 describe the changes in levels of hormone **B** after day 14 of the menstrual cycle and explain how these changes relates to the function of hormone **B**.

....

....

....

....

....

.... [3]

For Examiner's Use

(c) (i) On figure 4.2 **draw in** and **label** a line to represent the changes in levels of one other hormone involved in the menstrual cycle. [1]

(ii) Name **any** method of birth control that prevents the spread of Sexually Transmitted Infections (STIs) and explain how this method prevents pregnancy.

....

....

....

.... [2]

[13]

For Examiner's Use

5 Excretion is the process of removing metabolic wastes and excess substances from the body.

(a) Name the main product of metabolism excreted by the human kidney.

.... [1]

(b) Name **two** substances that are re-absorbed into the blood in the kidney.

1 ....

2 .... [2]

For Examiner's Use

(c) Dialysis can be used to treat people whose kidneys do not function properly. Fig. 5.1 shows dialysis machine carrying out dialysis treatment.

Key

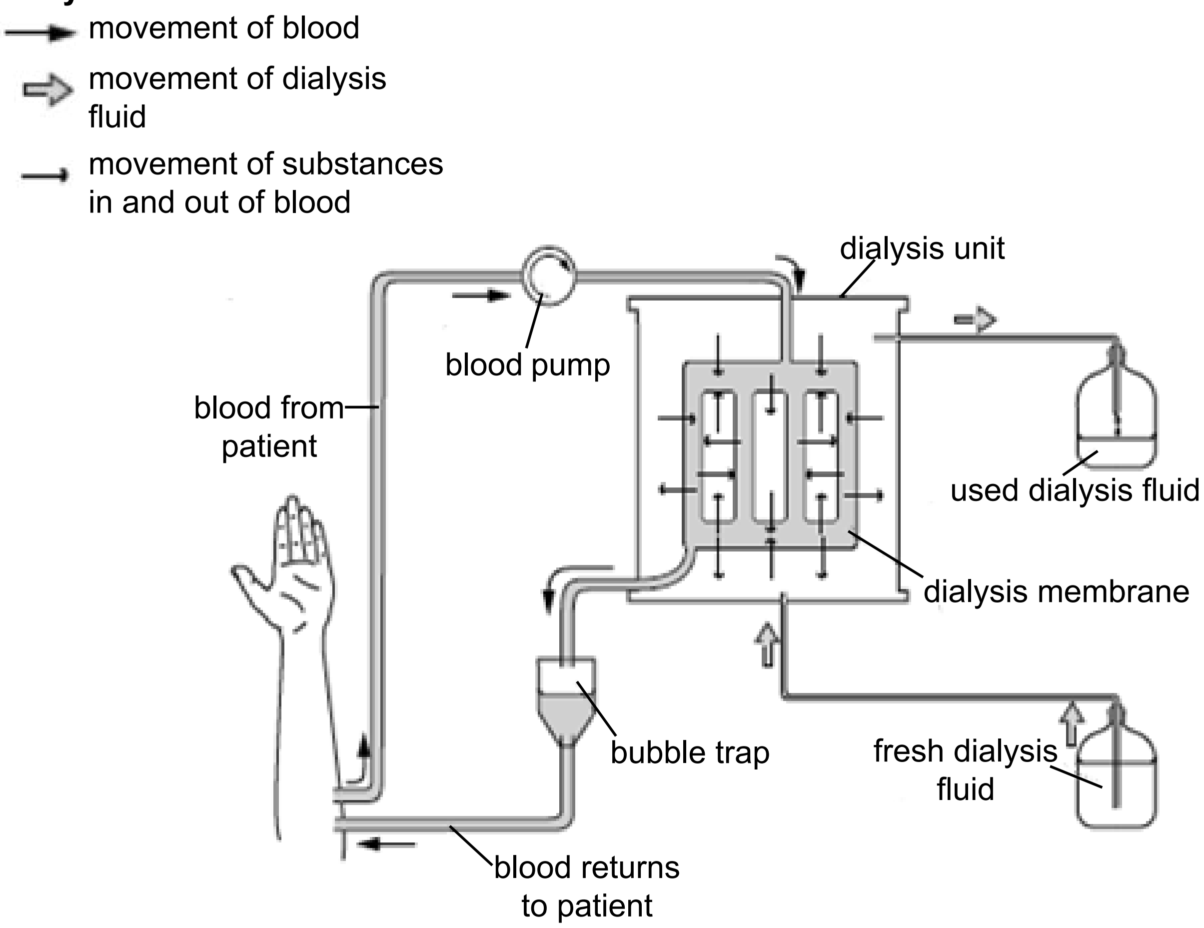


Fig. 5.1

(i) Explain the changes which occur to the blood as it goes through a dialysis machine.

....

....

....

....

....

.... [3]

For Examiner's Use

(ii) Some people with kidney failure are given a kidney transplant. Discuss advantages and disadvantages of a kidney transplant compared with dialysis.

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....

....

.... [4]

[10]

For Examiner's Use

6 Homeostasis enables animals to maintain a constant internal body temperature despite change in the environmental temperature.

(a) Describe the effects of vasodilation in maintaining human body temperature.

....

....

....

.... [2]

(b) Fig. 6.1 shows a summarised diagram of a negative feedback mechanism which controls homeostasis.

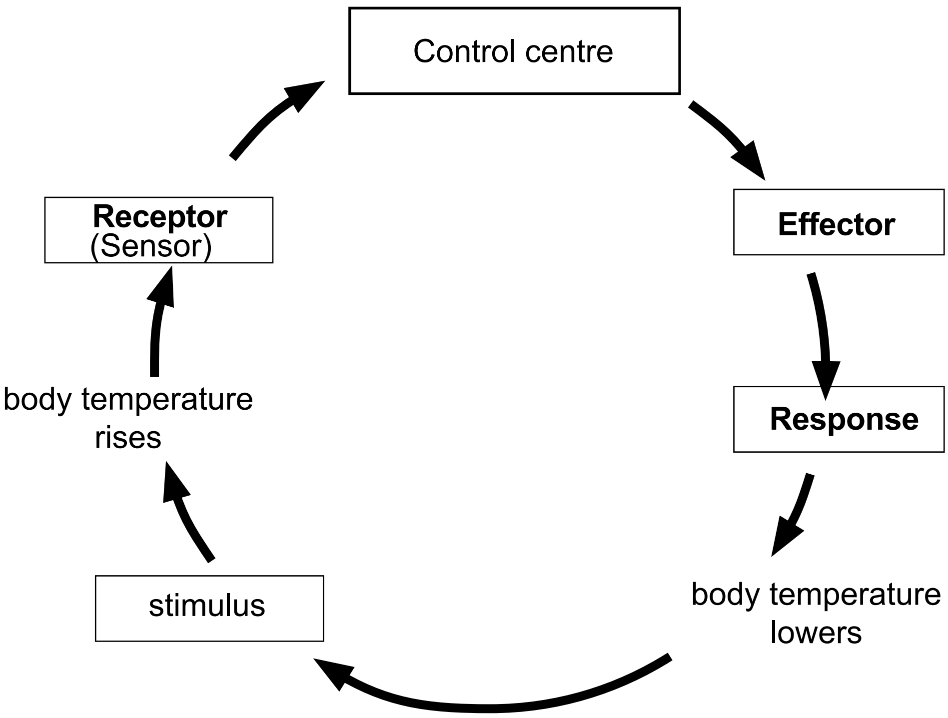

Fig. 5.1

(i) Explain the role of the following components during negative feedback mechanism.

Receptor (sensor) ....

.... [1]

Effector ....

.... [1]

Control centre ....

.... [1]

For Examiner's Use

(ii) The liver is an important organ in homoeostasis. Describe how the liver responds to an increase in glucose concentration of blood.

....

....

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.... [3]

[8]

For Examiner's Use

7 Fig. 7.1 shows an incomplete carbon cycle.

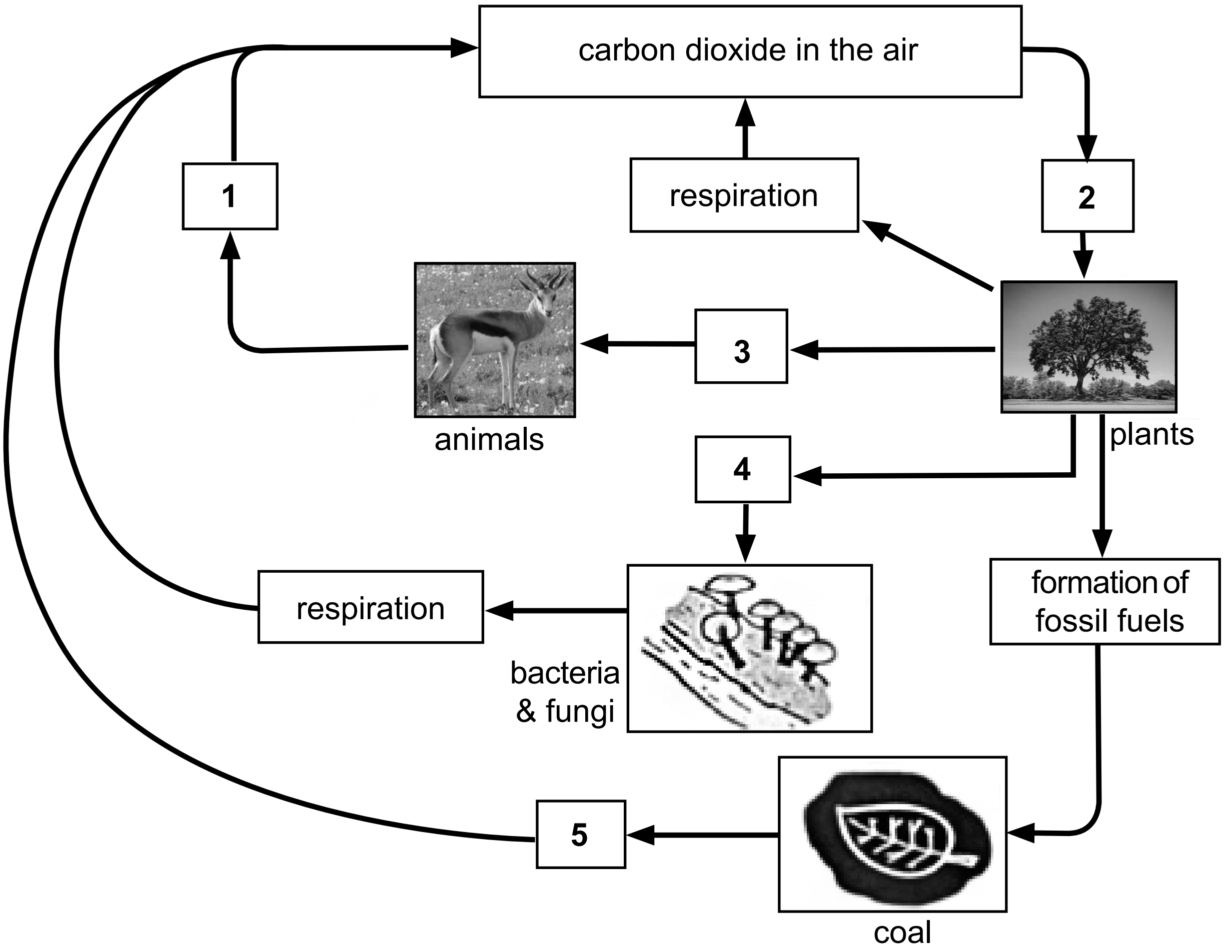


Fig. 6.1

(a) Name the processes taking place at point **1**,**2**,**3**,**4** and **5** in the cycle.

Write your answers in Table 7.1.

Table 7.1

Number	Name of process
1	....
2	....
3	....
4	....
5	....

[5]

For Examiner's Use

(b) Discuss any **two** effects of a long-term increase in the percentage of carbon dioxide in the atmosphere.

....

....

....

.... [2]

(c) Explain the effects of cutting down a forest on the balance between oxygen and carbon dioxide in the atmosphere.

....

....

....

.... [2]

(d) Explain how producers obtain the carbon they need to produce sugar (glucose).

....

....

....

.... [2]

(e) State the word equation for the production of sugar in plants.

....

....

....

.... [2]

[13]

For Examiner's Use

8 The eye colour in humans is controlled by two alleles. Allele **B** for brown eyes is dominant over allele **b** for blue eyes.

(a) Define the term *heterozygous*.

....

.... [1]

(b) Write down the genotype of a blue-eyed person.

....

.... [1]

(c) Draw a fully labelled genetic diagram to show the possible phenotype of offsprings produced by a heterozygous brown-eyed woman and a blue-eyed man.

[5]

(d) The genotype of a brown-eyed parent is not known. Describe how a genetic cross could reveal the genotype.

....

....

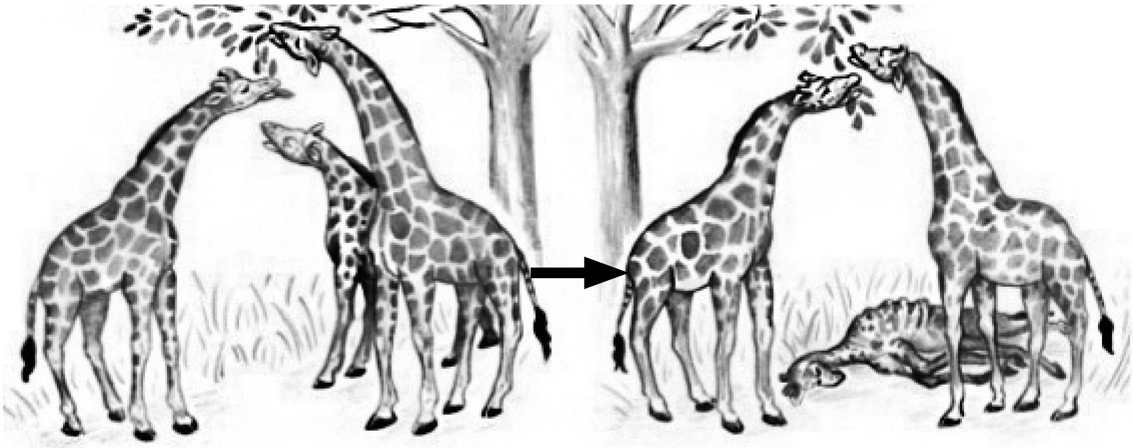
....

.... [2]

[9]

For Examiner's Use

9 Fig. 9.1 shows a population of giraffes feeding on a tall tree.



Population at the beginning   Same population over a period of time

Fig. 9.1

(a) Long-necked and short-necked giraffes co-exist in a population. After three generations, only long-necked giraffes and no short-necked giraffes were present in the population.

Referring to Fig. 9.1, suggest how the changes in the giraffe population may have come about.

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....

.... [4]

(b) Define *evolution*.

....

.... [1]

[5]

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